

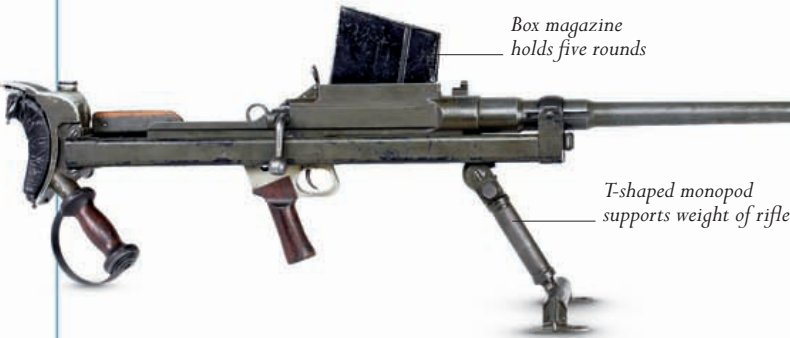
ANTI-TANK WEAPONS

The advent of the tank forced arms designers to come up with weapons powerful enough to counter this threat. For the infantry, who bore the brunt of the armoured assault, armour-piercing rifles came as a solution. Weapons such as the Panzerbüsche were also successful, but improvements in tank armour during World War II rendered them obsolete. Lightweight, short-range weapons firing shaped-charge projectiles became more useful. The most effective anti-tank weapon, however, was a high-velocity artillery piece.

▼ BOYS MK1 ANTI-TANK RIFLE

Date	1937
Origin	UK
Weight	16.33kg (36lb)
Barrel	91cm (35¾in)
Calibre	.55in
Armour penetration	21mm (¾in) at 302m (330 yards)

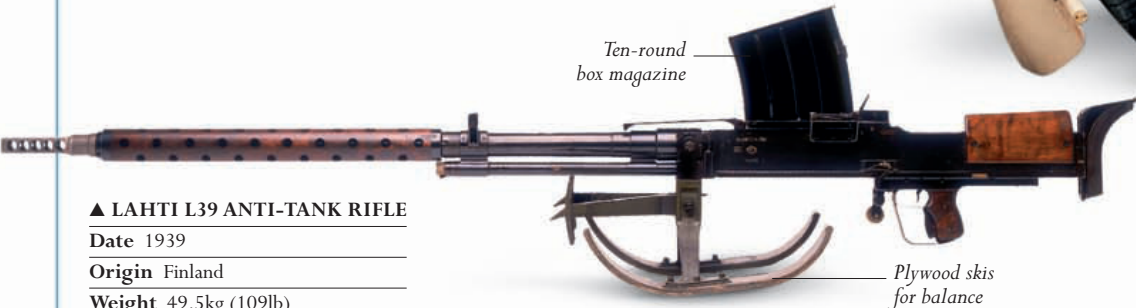
The Boys anti-tank rifle fired a heavy tungsten-steel round, and had a correspondingly violent recoil. However, it was only able to pierce light armour, and was replaced by the PIAT.



► FLAK 36 AA / AT GUN

Date	1936
Origin	Germany
Weight	7.4 tonnes (8.1 tons)
Length	5.79m (19ft)
Calibre	88mm
Armour penetration	159mm (6¼in) at 1,000m (1,094 yards)

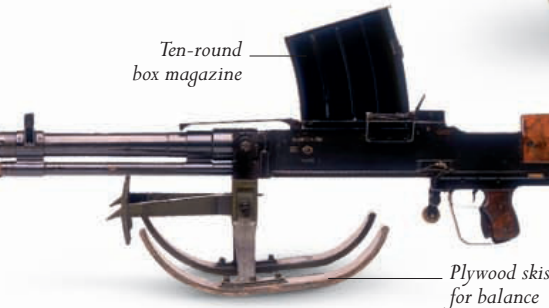
Designed as an anti-aircraft (AA) gun, the famed "88", as it was known, was found to be highly effective as an anti-tank gun. It could be put in position very quickly – within three minutes – although its bulk and height made it difficult to conceal. It was able to fire up to 20 rounds per minute.



▲ LAHTI L39 ANTI-TANK RIFLE

Date	1939
Origin	Finland
Weight	49.5kg (109lb)
Barrel	1.3m (4¼ft)
Calibre	20 × 138Bmm
Armour penetration	30mm (1¼in) at 100m (109 yards)

The L39's enormous size and weight gave it the nickname "Elephant Gun". It was used to good effect during the Winter War of 1939-40.



▼ PANZERBÜSCHE 39 ANTI-TANK RIFLE

Date	1940
Origin	Germany
Weight	12.6kg (27¾lb)
Barrel	1.08m (3½ft)
Calibre	7.92 × 94mm
Armour penetration	25mm (1in) at 300m (328 yards)

The Panzerbüsche 39 relied on its very high muzzle velocity and tungsten-cored bullet to penetrate enemy armour. It was, however, expensive to manufacture, and was only produced in small numbers.



◀ PAK 36 ANTI-TANK GUN

Date	1934
Origin	Germany
Weight	328kg (723lb)
Length	1.66m (5½ft)
Calibre	37mm
Armour penetration	38mm (1½in) at 365m (400 yards)

Designed for warfare in the 1930s, the light PAK 36 was obsolete by 1940. It was nicknamed the "doorknocker" for the way its shells bounced off the armour of Allied tanks.